Mustapha K. Khaldi

Oran, DZ | khaldi.mustapha.k@gmail.com | +213 562 98 91 09 | Google Scholar

Education

King Fahd University of Petroleum and Minerals, Dhahran, KSA
Ph.D in Systems & Control Engineering Sept 2021 – December 2025

Higher School of Applied Science, Tlemcen, DZ
Msc & Eng. Degrees in Systems & Control Engineering Jun, 2020

Experience

Adjunct Lecturer, University of Batna 2 - Mostefa Ben Boulaïd, Department of Mathematics Batna, DZ
September 2026 – Present

- Deliver Machine Learning and Deep Learning for Life Sciences course and its laboratory sessions;
- Prepare teaching materials and design homework assignments, quizzes, and examinations.

Manager, Essal Institute Oran, DZ
January 2026 – Present

- Led the design and rollout of technology training programmes, including summer camps, short professional courses, and industry-oriented curricula in AI, agentic AI, instrumentation, PLCs, and industrial automation.
- Recruited, interviewed, and coordinated instructors.
- Managed planning, budgets, enrollment coordination, and resource allocation for training activities.

Research Assistant, Interdisciplinary Research Center for Smart Mobility and Logistic, King Fahd University of Petroleum and Minerals Dhahran, KSA
Sept 2021 – December 2025

- Developed and evaluated data-driven models for crude-distillation and fluid-catalytic-cracking processes using Hammerstein-Wiener, NARX, neural-network, and Koopman-operator approaches in Python and MATLAB.
- Designed and studied advanced predictive-control strategies, including Koopman- and neural-network-based MPC and dual-mode predictive control for nonlinear industrial processes.
- Contributed to highway traffic forecasting research using Deep Koopman operator.
- Implemented and compared YOLOv5/YOLOv8 object detection for real-time logistics automation on NVIDIA Jetson devices, including GUI integration and cross-platform deployment.
- Develop and Implemented Control Strategies for drones (MPC, Neural Network MPC, Obstacle avoidance, Adaptive Control, Sliding mode Control).
- Supervise and mentor undergraduate students in "Self-Driving Car Student Competition"
- Wrote and edited manuscripts, Co-authored peer-reviewed surveys and reviews.

Teaching, Control and Instrumentation Engineering Department, King Fahd University of Petroleum and Minerals Dhahran, KSA
Sept 2021 – December 2025

Course Instructor

- CIE 318, Lab instructor Matlab / LabVIEW, troubleshooted and calibrated lab devices, developed new lab manual.
- CIE 305, Course instructor, graded assignments/exams, provided help sessions.

Teaching Assistant

- CIE 418/316, created case-study projects, supervised students, and graded assignments/exams.
- CIE 480/204, prepared teaching material slides and lecture notes, graded homework/assignments.

Research Assistant, SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum and Minerals Dhahran, KSA
Sept 2024 - May 2025

- Built an end-to-end scientific-keyword normalization pipeline combining syntactic and semantic analysis, fuzzy matching, SentenceTransformer embeddings, and LLM-based validation.
- Developed a web-based application for VOSviewer-compatible keyword co-occurrence analysis and visualization.

Instructor, Alfaisal University — Riyadh , KSA
Jul 2024 – Aug 2024

- Delivered hands-on instruction in AI, data literacy, block-based AI programming, Python, data visualization, and introductory machine learning for summer camp participants

Instructor, Essal Institute — Oran, DZ
Jun 2022 – Aug 2022

- Teaching Python level 1 and 2, Comprehensive course starting from the essentials to using libraries and automation scripts for efficient problem-solving
- Teaching Introduction to AI core concepts and foundations for building your own neural network.

Automation & SCADA Intern, SEOR (Oran Water Board), — Oran, DZ
Hassi Ben Okba Pumping Station — Feb 2020 – May 2020

- Developed a PLC-based automation stack for a 4 + 1 pump booster station, set up Profibus/industrial Ethernet topology and PLC-to-PLC data exchange between master/pump controllers.
- Station logic & interlocks: enforced tank-level constraints, suction/discharge thresholds, vibration/temperature trips; added hour-meter–based pump rotation, start/stop limits, and anti-chatter "temporary block" timers.
- Engineered water-hammer protection logic for upstream/downstream surge tanks, auto switchover between compressors, and shutdowns on persistent faults.
- Implemented backup power sequencing: transformer, genset, and coupling breaker logic with generator auto-start and alarms.
- Developed the HMI/SCADA in WinCC: main station overview, pump/valve detail, WHP screens, alarm classes, and trend views for flow/pressure/power

Publications

Bilinear Koopman Sliding-Mode Control with Disturbance Estimation for Robust Trajectory Tracking of Robotic Manipulators — Under review
Computers and Electrical Engineering

Data-Driven Adaptive Sliding Mode Control for Robot Manipulators via Koopman Operator Theory — Under review
Robotics and Autonomous Systems

Koopman-Based Dual-Mode MPC: Application to Fluid Catalytic Cracker - Fractionator process — Under review
Computers and Chemical Engineering

Forecasting of Highway Traffic Dynamics: a Deep Learning-Based Delay-Embedded Koopman Operator Approach — July 2026
10.1007/s13177-026-00704-y

Kinetics-based process design of integrated CO_2-to-Liquid Fuels and NO_x capture using desalination reject brine-derived sorbents — June 2026
10.1016/j.desal.2026.120451

CO_2-to-Liquid Fuels via C_2+ Electroreduction: Interplay of Kinetics, Catalyst Design, and Reactor Architecture — June 2026
10.1016/j.jece.2026.123350

Koopman-Lifted Dual-Mode Predictive Control for Disturbance Rejection in Industrial Processes — March 2026
10.1016/j.conengprac.2026.106918

Computational modeling of a Fluid Catalytic Cracking Unit — November 2025
10.1016/j.asej.2025.103661

Technical Aspects of Deploying UAV and Ground Robots for Intelligent Logistics Using YOLO on Embedded Systems — April 2025

10.3390/s25082572

From Block-Oriented Models to the Koopman Operator: A Comprehensive Review on Data-Driven Chemical Reactor Modeling — July 2025
10.3390/math13152411

Identification of Crude Distillation Unit: A Comparison between Neural Network and Koopman Operator — August 2024
10.3390/a17080368

Artificial intelligence perspectives: A systematic literature review on modeling, control, and optimization of fluid catalytic cracking — August 2023
10.1016/j.aej.2023.08.066

Optimal control of a three-stages supply chain model with shortages — May 2022
10.1109/SSD54932.2022.9955889

Supervision

Aremu Fawaz Babajide, Master Student (W/ Prof. Mujahed Al DHaifallah) — Dhahran, KSA
Multi-Objective Plant-Wide Optimization and Control of the Tennessee Eastman Process: From Mechanistic ENMPC to Surrogate-Assisted Data-Driven Real-Time Optimization — Spring and Fall 2025

Skills

Programming : Python, PyTorch, TensorFlow, MATLAB / Simulink, LabVIEW, Julia, VB.Net, PLC

Tools : Git, GitHub, CasADi, Acados, LaTeX, Excel, MS Word, MS PowerPoint

Languages : English, Arabic, French